

IMPERIAL

Group Action

Subheading

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Definition (Group Action)

A group action of a group G on a set X is a map $\cdot : G \times X \rightarrow X$ satisfying:

- (Compatibility) $(g_1 g_2) \cdot x = g_1 \cdot (g_2 \cdot x)$
- (Identity) $1 \cdot x = x$

```

2  /-! ## Definitions: group actions -/
3  /-- Defines a group action of a monoid `G` on a type `X`.
4  The action is given by `act : G → X → X`,
5  satisfying the axioms `ga_mul` and `ga_one`. -/
6  class GroupAction (G : Type*) [Monoid G] (X : Type*) where
7    /-- The action function that applies a group element to an element of `X`. -/
8    act : G → X → X
9    /-- The multiplication axiom: `(g * g) • x = g • (g • x)`. -/
10   ga_mul : g g x, act (g * g) x = act g (act g x)
11   /-- The identity axiom: `1 • x = x`. -/
12   ga_one : x, act 1 x = x

```

Faithful Action

An action is faithful if distinct group elements induce distinct permutations:

$$(\forall x \in X, g_1 \cdot x = g_2 \cdot x) \implies g_1 = g_2$$

```
5  /-! ## Definitions: act faithfully -/
6  /-- A group action is faithful if different group
   ⇨ elements
```

Transitive Action

An action is transitive if any two elements can be connected:

$$\forall x_1, x_2 \in X, \exists g \in G \text{ s.t. } g \cdot x_1 = x_2$$

```
33  /-! ## Definitions: transitive -/
34  /-- A group action is transitive if for any `x , x
   ⇨ `X`,
35  there exists a group element `g` `G` such that `g
   ⇨ • x = x`. -/
36  def GroupAction.transitive {G : Type*} [Group
   ⇨ G] {X : Type*} [GroupAction G X] : Prop :=
37      x x : X, g : G, GroupAction.act g x = x
38
39  #lint
```

For each $g \in G$, define $\sigma_g : X \rightarrow X$ by $\sigma_g(x) = g \cdot x$. This yields the permutation representation $\phi : G \rightarrow \text{Sym}(X)$.

For $x \in X$, the stabilizer is $G_x = \{g \in G \mid g \cdot x = x\}$.

Theorem: G_x is a subgroup of G .

The symmetric group $\text{Sym}(X)$ acts on X by evaluation: $\sigma \cdot x := \sigma(x)$.